

First Law of Thermodynamics

2. Fundamental Concept

System: A system can be any object, region of space etc, selected for study and set apart (mentally) from everything else. Which then become surrounding.

The system of interest in thermodynamics are finite and macroscopic rather than microscopic. The imaginary envelope which encloses a system and separated it from its surrounding is called the boundary of the system.

Isolated system: It can not exchange either matter or energy with the surroundings.

If exchange of matter is allowed the system is said to be open; if only energy and not matter is closed (but not isolated).

Thermodynamical state: A thermodynamic state is a set of values of properties of a thermodynamic system that must be specified to reproduce the system.

Thermodynamic state is the macroscopic condition of a thermodynamic system as described by its particular thermodynamic parameter. Such as temperature (T) pressure (P) volume (V) density (ρ).

State function: state function also called “State variable” thermodynamic variables describe the momentary condition of thermodynamic system

For a continuous process, such variable are exact different also fully determined by their initial and final thermodynamic states.

Example includes entropy, pressure, temperature, volume, etc.

Intensive and extensive properties: Intensive properties:-it is a physical property of a system that does not depend on the system size or the amount of material in system. It is scale invariant.

Example: Chemical potential, density, viscosity, resistivity, specific heat capacity, pressure, elasticity, magnetization, velocity, acceleration, temperatures, etc.

Solution: Extensive properties:- It is one that is additive for independent non-interacting subsystem. It is directly proportional to the amount of material in the system.

Example energy, Entropy, Gibbs energy, mass momentum, volume, change, weight,

Note: If f and g are arbitrary intensive variable then $fg, \frac{f}{g}, \frac{df}{dg}$ and $f + g$ will also intensive variable.

If F and G are also two arbitrary extensive variable then $F + G$ will an extensive variable but $\frac{F}{G}$ and $\frac{dF}{dG}$ will be intensive.

If F is extensive variable and f is intensive variable the fF and $\frac{F}{f}$ and $\frac{dF}{df}$ is extensive variable.